

Quantum Computation and Information

Lecture 1: Basics of Quantum Mechanics

- course logistics
- basics of quantum mechanics

course logistics

basic information

- canvas: <https://canvas.colorado.edu/courses/143314>
- time and location: TTh 2:00-3:15pm, ECCR 210
- office hours:
 - Tue 4:00pm-5:00pm at ECCR 1B12
 - by appointment (email is most convenient)

- no required textbook, but some suggested readings
 - see canvas for useful lecture notes and text for quantum computing
- prerequisite: linear algebra

grading and project

- grading: 30% homework, 30% midterm exam, 30% final projects, 10% attendance
- homework:
 - 6 homework sets
 - due by midnight on Tuesdays
 - collaboration strongly encouraged, but write your own solutions
- final projects: more detail later in the semester

bit versus qubit

bit versus qubit

bit

- basic unit of classical information that takes a value of **either 0 or 1**
- a two-state classical system: on or off, low or high, . . .

quantum bit (qubit)

- basic unit of quantum information that takes a value in two-dimensional complex vector space

$$|\psi\rangle = a|0\rangle + b|1\rangle, \quad |a|^2 + |b|^2 = 1$$

- $|0\rangle$ and $|1\rangle$: elements of an orthonormal basis
 - $a, b \in \mathbb{C}$: the probability amplitudes
 - if measured onto the basis, $|0\rangle$ is obtained with probability $|a|^2$ and $|1\rangle$ with probability $|b|^2$
- a two-dimensional quantum system with two distinguishable states

basics of quantum mechanics

- how to represent a quantum state?
- how to represent an observable on quantum system?
- how does a quantum system evolve in time?
- what operations can be applied to a quantum system?
- how to combine multiple quantum systems into a composite system?

basics of quantum mechanics

I: quantum state

- a unit vector in Hilbert space $\mathcal{H}$, denoted by $|\psi\rangle$ (Dirac's ket notation)
 - a vector space over complex numbers $\mathbb{C}$
 - inner product $\langle\psi|\varphi\rangle = \langle\varphi|\psi\rangle^*$ mapping an ordered pair of vectors to $\mathbb{C}$
 - dual vector $\langle\psi|$ (Dirac's bra notation) that maps vectors to complex numbers
 - $\|\psi\| = \sqrt{\langle\psi|\psi\rangle} = 1$
 - $|\psi\rangle$ and $e^{i\alpha}|\psi\rangle$, $\alpha \in \mathbb{R}$ are indistinguishable

- a quantum system with N distinguishable states $|0\rangle, |1\rangle, \dots, |N-1\rangle$

$$|\psi\rangle = a_0 |0\rangle + a_1 |1\rangle + \dots + a_{N-1} |N-1\rangle, \quad \sum_{i=0}^{N-1} |a_i|^2 = 1$$

- a superposition of N states, with corresponding amplitudes a_i
- regarding distinguishable states as orthonormal basis of $\mathcal{H} = \mathbb{C}^N$

$$|\psi\rangle = \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_{N-1} \end{pmatrix}$$

- conjugate transpose $\langle\psi| = (a_0^*, a_1^*, \dots, a_{N-1}^*)$
- for two states $|\psi\rangle = a_0 |0\rangle + a_1 |1\rangle + \dots + a_{N-1} |N-1\rangle$ and $|\varphi\rangle = b_0 |0\rangle + b_1 |1\rangle + \dots + b_{N-1} |N-1\rangle$, the inner product

$$\langle\psi|\varphi\rangle = a_0^* b_0 + a_1^* b_1 + \dots + a_{N-1}^* b_{N-1}$$

- qubit: $\mathcal{H} = \text{span}\{|0\rangle, |1\rangle\}$, state $|\psi\rangle = a |0\rangle + b |1\rangle$ with $|a|^2 + |b|^2 = 1$
- first hint to the power of QC: **quantum parallel**

II: observable

- a property of physical system that can be measured
- mathematically a self-adjoint operator on Hilbert space

$$A = A^\dagger$$

– example: $\begin{pmatrix} 3 & i \\ -i & 1 \end{pmatrix}$

- can be diagonalized

$$A = \sum_n \alpha_n E_n$$

- $\alpha_n \in \mathbb{R}$: eigenvalue of A
- $E_n = |n\rangle \langle n|$: the orthogonal projector onto the space of eigenvector/eigenstate $|n\rangle$ with eigenvalue α_n

$$\begin{aligned} E_n &= E_n^\dagger \\ E_n E_m &= \delta_{n,m} E_n \end{aligned}$$

III: measurement

- the measurement of an observable A prepares an eigenstate $|n\rangle$ of A (*state collapse*), obtains the value of the corresponding eigenvalue α_n
- given quantum state $|\psi\rangle$, the probability to obtain outcome α_n

$$\text{Prob}(\alpha_n) = \|E_n |\psi\rangle\|^2 = \langle\psi| E_n |\psi\rangle$$

- if outcome α_n is obtained, the state after measurement $\frac{E_n |\psi\rangle}{\|E_n |\psi\rangle\|}$
 - if measured again right after, the same outcome is obtained
- expected value of measurement outcome on many identical quantum systems

$$\sum_n \alpha_n \text{Prob}(\alpha_n) = \sum_n \alpha_n \langle\psi| E_n |\psi\rangle = \langle\psi| A |\psi\rangle$$

IV: dynamics

- evolution of a (closed) quantum system from time t to t' is governed by a unitary operator

$$|\psi(t')\rangle = U(t', t) |\psi(t)\rangle$$

- unitary operator satisfying $U^\dagger U = U U^\dagger = I$
- example: $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$
- unitary operator preserves inner product: given $|\psi'\rangle = U |\psi\rangle$ and $|\varphi'\rangle = U |\varphi\rangle$, $\langle\psi'|\varphi'\rangle = \langle\psi| U^\dagger U |\varphi\rangle = \langle\psi|\varphi\rangle$

- infinitesimal time evolution is governed by *Schrodinger* equation

$$\frac{d}{dt} |\psi(t)\rangle = -iH(t) |\psi(t)\rangle$$

- Hamiltonian H is self-adjoint

- two type of operation on quantum system: **unitary transformation** and **measurement**

V: composite system

- given systems $\mathcal{A}$ with Hilbert space $\mathcal{H}_{\mathcal{A}}$ and $\mathcal{B}$ with $\mathcal{H}_{\mathcal{B}}$, Hilbert space of composite system $\mathcal{AB}$ is the tensor product of $\mathcal{A}$ and $\mathcal{B}$

$$\mathcal{H}_{\mathcal{AB}} = \mathcal{H}_{\mathcal{A}} \otimes \mathcal{H}_{\mathcal{B}}$$

- with dimension $d_{\mathcal{A}}d_{\mathcal{B}}$ and orthonormal basis $\{|i\rangle_{\mathcal{A}} \otimes |j\rangle_{\mathcal{B}}\}$
- the state of composite system is the product of the states of subsystems $|\psi\rangle_{\mathcal{AB}} = |\psi\rangle_{\mathcal{A}} \otimes |\psi\rangle_{\mathcal{B}}$

- example: a system of two qubits

- dimension: 4; orthonormal basis: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$
- given states of two qubits $a|0\rangle + b|1\rangle$ and $c|0\rangle + d|1\rangle$, the state of composite system is

$$\begin{aligned} & (a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) \\ = & ac|0\rangle \otimes |0\rangle + ad|0\rangle \otimes |1\rangle + bc|1\rangle \otimes |0\rangle + bd|1\rangle \otimes |1\rangle \\ = & ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle \end{aligned}$$

- second hint to the power of QC: **exponential representation capacity**

entanglement

- state of two-qubit system is a vector in 4-dimensional Hilbert space

- orthonormal basis: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$

- separable bipartite system: its state can be written as the tensor product of two states

- example

$$\frac{1}{2}|00\rangle - \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle - \frac{1}{2}|11\rangle = \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) \otimes \left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right)$$

- each party has its 'own' state

- entangled state: state is not separable

- example

$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle \quad (\text{Bell state, or EPR pair})$$

- Alice and Bob are physically separate, each with one qubit. what happens if Alice measures her qubit, say, onto the orthonormal basis?
 - separable state $\frac{1}{2} |00\rangle - \frac{1}{2} |01\rangle + \frac{1}{2} |10\rangle - \frac{1}{2} |11\rangle$?
 - * Alice observes $|0\rangle$ ($|1\rangle$) with probability $1/2$ ($1/2$)
 - * composite system state $\frac{1}{\sqrt{2}} |00\rangle - \frac{1}{\sqrt{2}} |01\rangle$ ($\frac{1}{\sqrt{2}} |10\rangle - \frac{1}{\sqrt{2}} |11\rangle$)
 - * Bob's qubit state $\frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle$, independent of Alice's observation
 - * what Alice does to her qubit has no effect on Bob's
 - entangled state $\frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle$?
 - * Alice observes $|0\rangle$ ($|1\rangle$) with probability $1/2$ ($1/2$)
 - * composite system state $|00\rangle$ ($|11\rangle$)
 - * Bob's qubit state $|0\rangle$ ($|1\rangle$), depending on Alice's measurement

teleportation?

- Alice and Bob are physically separate, each with one qubit, and share a Bell state. What happens if Alice measures her qubit, say, onto the orthonormal basis?

$$\frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle$$

- if Alice measures onto $|0\rangle$, she instantly knows Bob would be in state $|0\rangle$; and if Bob measures on the same basis, he will find $|0\rangle$
- if Alice measures onto $|1\rangle$, she instantly knows Bob would be in state $|1\rangle$; and if Bob measures on the same basis, he will find $|1\rangle$

teleportation?

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- if Alice measures onto $|0\rangle$, she instantly knows Bob would be in state $|0\rangle$; and if Bob measures on the same basis, he will find $|0\rangle$
- if Alice measures onto $|1\rangle$, she instantly knows Bob would be in state $|1\rangle$; and if Bob measures on the same basis, he will find $|1\rangle$
- does this convey any information from Alice to Bob?

teleportation?

- Alice and Bob are physically separate, each with one qubit, and share a Bell state. What happens if Alice measures her qubit, say, onto the orthonormal basis?

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- if Alice measures onto $|0\rangle$, she instantly knows Bob would be in state $|0\rangle$; and if Bob measures on the same basis, he will find $|0\rangle$
- if Alice measures onto $|1\rangle$, she instantly knows Bob would be in state $|1\rangle$; and if Bob measures on the same basis, he will find $|1\rangle$
- does this convey any information from Alice to Bob?
 - * no, they just acquire correlated random bits; Bob does not know if his measurement outcome is conditioned on Alice's measurement
 - * no communication theorem: nothing that Alice does changes Bob's local state, in terms of measurement only on his qubit
- however, Bell state can be leveraged to transport quantum information/state between two distant parties (quantum teleportation)

density matrix

- Alice and Bob, physically separated, share an entangled state $|\psi\rangle = a|00\rangle + b|11\rangle$, $|a|^2 + |b|^2 = 1$. Can we have a description of, say, Alice's qubit alone?
- observable on Alice's only $\mathcal{M}_A \otimes I_B$, with expected value

$$\begin{aligned}\langle\psi| \mathcal{M}_A \otimes I_B |\psi\rangle &= (a^* \langle 00| + b^* \langle 11|)(\mathcal{M}_A \otimes I_B)(a|00\rangle + b|11\rangle) \\ &= |a|^2 \langle 0| \mathcal{M}_A |0\rangle + |b|^2 \langle 1| \mathcal{M}_A |1\rangle \\ &= \text{tr}(\mathcal{M}_A \rho_A)\end{aligned}$$

- density matrix/operator $\rho_A = |a|^2 |0\rangle \langle 0| + |b|^2 |1\rangle \langle 1|$
 - * for Bell state $\rho_A = \frac{1}{2} |0\rangle \langle 0| + \frac{1}{2} |1\rangle \langle 1|$
- Alice is in mixed state: $|0\rangle$ with probability $|a|^2$ and $|1\rangle$ with probability $|b|^2$
- it is a 'complete' description of Alice's as it's consistent with any measurement on Alice's only

density matrix

- given (pure) quantum state $|\psi\rangle$, density matrix $\rho = |\psi\rangle \langle\psi|$, expected value of observable $\mathcal{M}$

$$\langle\psi| \mathcal{M} |\psi\rangle = \text{tr}(\mathcal{M}\rho)$$

- bipartite system $\mathcal{H}_A \otimes \mathcal{H}_B$ with state

$$|\psi\rangle = \sum_{i,\mu} a_{i\mu} |i\rangle_A \otimes |\mu\rangle_B = \sum_{i,\mu} a_{i\mu} |i\rangle_A |\mu\rangle_B, \quad \sum_{i,\mu} |a_{i\mu}|^2 = 1$$

expected value of observable $\mathcal{M}_A \otimes I_B$ acting only on subsystem A

$$\begin{aligned} \langle\psi| \mathcal{M}_A \otimes I_B |\psi\rangle &= (\sum_{i,\mu} a_{i\mu}^* \langle i|_A \langle \mu|_B) \mathcal{M}_A \otimes I_B (\sum_{j,\nu} a_{j\nu} |j\rangle_A |\nu\rangle_B) \\ &= \sum_{i,j,\mu} a_{i\mu}^* a_{j\mu} \langle i|_A \mathcal{M}_A |j\rangle_A \\ &= \text{tr}(\mathcal{M}_A \rho_A) \end{aligned}$$

– density matrix $\rho_A = \text{tr}_B |\psi\rangle \langle\psi| = \sum_{i,j,\mu} a_{i\mu}^* a_{j\mu} |j\rangle_A \langle i|_A$

density matrix

- ρ is self-adjoint, i.e., $\rho = \rho^\dagger$
- ρ is positive, i.e., $\langle \varphi | \rho | \varphi \rangle \geq 0$ for any $|\varphi\rangle$
- $\text{tr } \rho = 1$
- pure state if $\text{rank}(\rho) = 1$ and mixed state otherwise
- pure state if $\rho^2 = \rho$ and mixed state otherwise

no-cloning theorem

no unitary transformation can make a copy of an arbitrary quantum state

- take two arbitrary states $|\psi\rangle_A$ and $|\varphi\rangle_A$

$$\begin{aligned}U |\psi\rangle_A |0\rangle_E &= |\psi\rangle_A |\psi\rangle_E \\U |\varphi\rangle_A |0\rangle_E &= |\varphi\rangle_A |\varphi\rangle_E\end{aligned}$$

- unitary transformation preserves inner product

$$\langle\psi|\varphi\rangle = \langle\psi|\varphi\rangle^2$$

- $\langle\psi|\varphi\rangle = 0$ (orthogonal) or $\langle\psi|\varphi\rangle = 1$ (identical), a contradiction